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Palaeomagnetic resetting in the Barrovian zones of Scotland and its relationship to the late structural history

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The haematite schists in the lowest kyanite and the staurolite Barrovian zones exposed in Glen Lethnot and Glen Esk, Angus, Scotland, have a characteristic magnetization with blocking temperatures generally greater than 600°C and less than 695°C and coercivity greater than 200 mT. The mean palaeomagnetic direction derived from 54 samples collected from $N = 10$ sites is $D = 159°$, $I = 45°$, $k = 47$, $\alpha_{95} = 7.1°$ with the associated south pole position of lat. $= -3°$, long. $= 22°$E, $dp = 5.7°$ and $dm = 9.0°$. This magnetization is younger than folds of D3 generation and older than folds of D4 generation. The pole position falls on the Aberdeenshire polar wander track in a place consistent with an average K-Ar biotite chrontour age of 455 m.y. A comparison of the blocking temperatures of magnetization to the estimated peak temperature of metamorphism suggest the magnetization is a chemical remanent magnetization (CRM) possibly acquired during the metamorphic chemical reactions.

1. Introduction

With the emergence of a significant amount of palaeomagnetic data derived from stepwise demagnetization, it is now feasible to investigate the palaeomagnetic record of metamorphic terrains to obtain information on the thermal and structural history. Palaeomagnetic data are available from Precambrian metamorphic terrains but these studies generally emphasize the construction of portions of polar wander paths. In this paper I present palaeomagnetic results from haematite schists in the staurolite and kyanite zones of Glen Esk and Glen Lethnot, Angus, Scotland, within the area where Barrow [1,2] demonstrated the existence of zones of progressive regional metamorphism.

The objectives of this investigation are: (1) to identify characteristic magnetizations acquired during metamorphic recrystallization and/or during uplift and cooling of the metamorphic terrain; (2) to determine the timing of magnetization relative to different generations of minor structures; (3) to compare the magnetizations with those from other parts of the Dalradian and to relate them to the isotopic cooling age pattern [3,4]. If magnetizations of purely thermal origin can be identified, it may be possible to estimate temperatures and cooling rates at the time of magnetization [5]. If the magnetization is of chemical origin, it may have been acquired during the peak of metamorphic recrystallization.

2. Geology of the haematite schists

Fig. 1, after Harte and Johnson [6] shows the sampling sites with respect to the metamorphic isograds in the Dalradian pelitic assemblages. In the Dalradian, four phases of structural deformation are generally recognized [7] and a fifth is developed locally [6]. D1 and D2 correspond to the principal and later stages of nappe emplacement. In Glen Esk and Glen Lethnot, D3 folds the earlier schistosity into tight, asymmetrical herring-bone folds with near horizontal axial planes. The trend of the D4 fold axis ranges from E-W to NE-SW [6]. These folds have a distinct axial plane cleavage. The latest structures are monoclines and kink bands that fold the earlier

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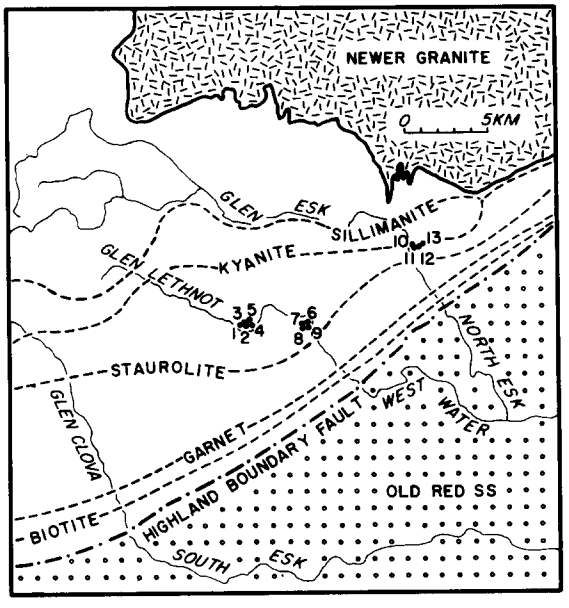


Fig. 1. Geological map after [6] showing palaeomagnetic sampling sites with respect to the metamorphic isograds in the Barrovian zones.

schistosities. Harte and Johnson [6] locally recognize D4 and D5 late structures, distinguished by the orientation of the axial plane. D4 planes strike NE-SW (parallel to the Highland Boundary Fault) and dip to the northwest. D5 planes strike WNW-ESE. A large, late, dome structure in Glen Esk, the Tarfside Culmination, is recognised as D5 in generation [6].

Observing the relations of metamorphic mineral growth to structural fabric, Harte and Johnson [6] conclude that the peak of metamorphic temperatures occurred either during or shortly after D3 and that a considerable cooling from the peak temperature took place between D3 and D4. The peak temperatures in the staurolite and lowest kyanite zones were probably about 580–600°C at a pressure of 6 kbar [8]. Retrogressive metamorphism may date from the period of D4 folding. Concentrations of chlorite are commonly found along the D4 axial fold planes. Usually the amount of retrogression is very limited though virtually all rocks show a small amount of chlorite forming from biotite.

Haematite schists are found in the higher-grade zones in Glen Esk and Glen Lethnot. In many places they are associated with porphyroblasts of kyanite and staurolite. The main primary ore is haematite and may be accompanied by ilmenite, magnetite, pyrite or rutile. Magnetite sometimes shows secondary alteration to haematite but the greatest portion of the haematite is primary and forms tabulae usually lying parallel to the schistosity (B. Harte, personal communication). These schists may constitute an original stratigraphic horizon (B. Harte, personal communication).

Dewey and Pankhurst [3] have summarized the geochronological data for the Scottish Highlands in relation to the metamorphic and cooling history. The published K-Ar age determinations from the Barrovian zones in the vicinity of the palaeomagnetic sampling sites are sparse with two biotite ages from the kyanite zone of 452 m.y. and 455 m.y. (recalculated from [9] and [10] using standard decay constants [11]) and one from the upper garnet zone of 457 m.y. (recalculated from [9]).

Dempster [4] has carried out Rb-Sr age determinations on biotites and muscovites separated from rocks collected on a traverse of the Barrovian zones, beginning in the sillimanite zone and ending in the staurolite zone. These data show that the kyanite and staurolite zones covered by the traverse cooled from approximately 500°C to 300°C during the period 460 m.y. to 450 m.y. ago.

3. Properties of the remanent magnetization

Oriented drill cores 2.54 cm in diameter were collected from the thirteen sites shown in Fig. 1. Orientations were measured in situ using a clinometer and a magnetic compass. The magnetic bearings were checked when possible using a solar compass. The cores were sliced into cylinders 2.3 cm high.

Magnetic moments were measured with a complete results fluxgate magnetometer [12]. Thermal demagnetization was carried out using an oven in a field free space maintained by orthogonal coils with a feedback system [13,14]. Alternating field demagnetization up to peak fields of 200 mT was carried out using a tumbler device [15].

The schists have natural remanent magnetization (NRM), intensities ranging from 0.74 to 25.7 A/m. Fig. 2 shows orthogonal projections [16] of the results of some demagnetization experiments, the most typical of which are illustrated in

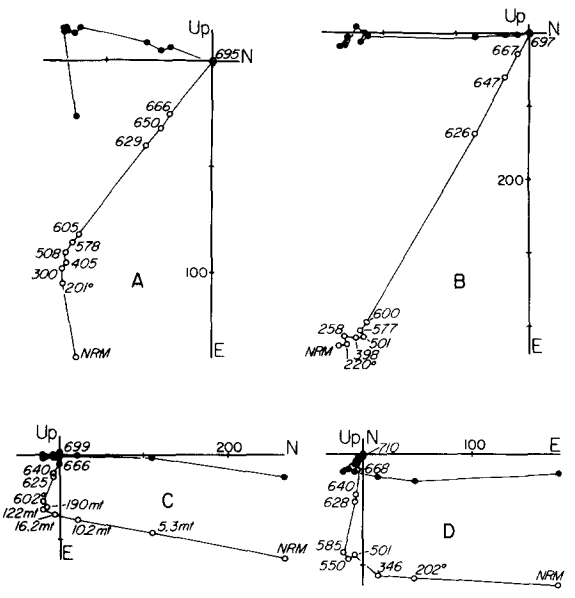


Fig. 2. Orthogonal projections [16] showing typical demagnetization trajectories of the haematite schists in the Barrovian zones. Units of the axes are $\times 10^{-3}$ A/m. Open circles denote projections on the vertical plane and closed circles denote projections on the horizontal plane. Numbers adjacent to data points denote demagnetization temperatures (A, B, D) or peak alternating field in mT (C).

2A and 2B. The characteristic magnetization has blocking temperatures greater than 600°C, and less than 695°C, and remanent coercivities greater than 200 mT. In some samples, this is overprinted by a secondary component. In the examples illustrated in 2A and 2B this secondary component is small and has low blocking temperatures; in other instances, as illustrated in 2C and 2D, it has blocking temperatures up to 585°C and remanent coercivites up to 16 mT. The secondary magnetizations have random orientations and no explanation of their origin is offered. The directions of the characteristic magnetization used to compute the mean were determined using the principal component analysis algorithm of Kent et al. [17].

4. Relationship of the characteristic magnetization to D3 and D4 folds

The timing of the acquisition of the characteristic magnetization can be determined by applying the fold test [18] to folds of different generation. If the peak metamorphic temperature was reached during or shortly after D3 and a substantial drop of temperature occurred between D3 and D4, the characteristic magnetization would pre-data D3 but post-date D4 in the higher-grade zones. If the characteristic magnetization pre-dates D3, then the distribution of directions of magnetization would be chaotic in areas where D3 folds are well developed because of the complex and intense nature of D3 folding. Low-temperature, post-metamorphic chemical remanent magnetization (CRM) would post-date D4.

Within site 11 is a very tight D3 herring-bone fold with a subhorizontal axial plane and a fold axis plunging gently (~ 8°) to the northeast (Fig. 3). The lower limb of the fold is overturned with respect to the upper limb. Six oriented cores were collected from this fold; three from the upper limb and three from the lower limb. As shown in Fig. 3, the characteristic magnetization is not folded by this structure.

The outcrop at site 8 in Glen Lethnot exhibited small-scale, chaotic changes in the attitude of the regional fabric, due to folding of D3 generation. The directions of the characteristic magnetization are not affected by the orientation of the fabric at

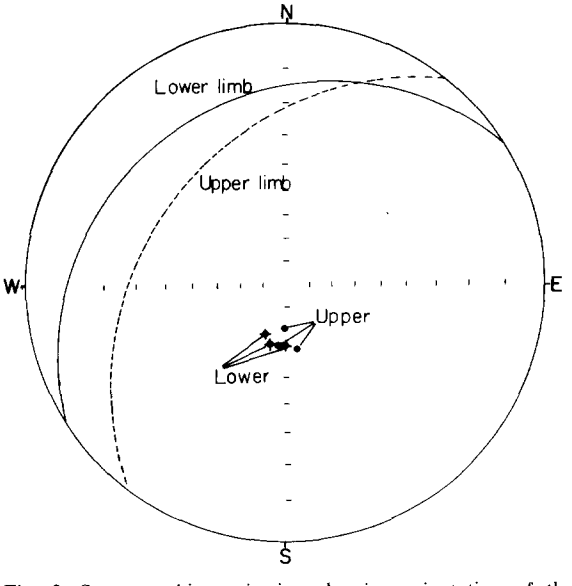


Fig. 3. Stereographic projection showing orientation of the upper (dashed to show intersection with upper hemisphere) and lower (solid to show intersection with lower hemisphere) limbs of D3 fold at site 11, and the directions of characteristic magnetization in samples from the upper (solid dots) and lower (solid dots with cross) limb of the fold.

this site. The estimate of the precision parameter, k, is 88 (Table 1) and is comparable to sites which exhibit no evidence of D3 folding. The evidence from these two sites shows that the characteristic magnetization post-dates D3 folding in this part of the Barrovian zones.

Three folds of D4 generation were sampled; two from site 1 and one from site 3 in Glen Lethnot. One oriented core was collected from the steep limb of the two folds in site 1 and two cores were collected from the steep limb of the fold at site 3. In each case, the size of the fold permitted only this limited sampling. Additional samples were collected from parts of the outcrops where the regional fabric is not affected by D3 or D4 folding.

In Glen Esk, sites 10, 11 and 12 are probably affected by a D4 structure as the average dip of the regional fabric (about 45°) is substantially steeper than in most parts of the Barrovian zones in Glen Esk (B. Harte, personal communication). Fig. 4A shows the in-situ directions of the characteristic magnetization derived from the steep limb of the D4 monoclines at sites 1 and 3, the site-mean directions from sites 10, 11 and 12 and the mean direction of magnetization determined from sites not affected by D4 deformation. Fig. 4B shows these directions after structural correction is applied. In the case of the small-scale D4 folds, the palaeomagnetic directions are rotated about the average strike of the fabric by the angle required to restore the fabric to the same attitude as the regional fabric. The correction to sites 10, 11 and 12 is about the strike of the fabric by the angle required to restore the orientation of the fabric to the attitude generally observed in the higher grade

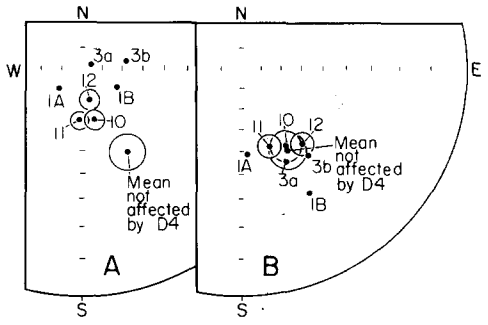


Fig. 4. Stereographic projections showing the direction of characteristic magnetization of samples collected from steep limbs of D4 folds before (A) and after (B) structural correction compared to the mean direction (Table 1) of magnetization from sites not affected by D4 deformation. 1A and 1B are single samples from two separate D4 folds within site 1, and 3a and 3b are two samples from the same D4 fold within site 3.

TABLE 1

Mean directions, poles and statistics

Site No.	Grid ref.	N	$\bar{D}$(°)	$\bar{I}$(°)	k	α_{95}(°)	Pole (°N, °E)	dp(°)	dm(°)
1 a	N0465720	7	168	54	77	7	2, 7	6.9	9.8
2	N0467720	4	150	32	63	12	−12, 26	7.6	13.5
3 a	N0467721	3	154	34	223	8	−11, 22	5.2	9.1
4	N0468721	5	151	40	53	11	−6, 24	8.0	13.2
5	N0468722	4	171	44	52	13	−7, 5	10.2	16.3
6	N0505723	5	142	46	33	13	0, 30	10.6	16.6
7	N0502723	5	139	56	256	5	9, 30	5.2	7.2
8	N0502723	5	159	39	88	8	−9, 17	5.7	9.5
9	N0505721	4	138	50	276	6	5, 32	5.4	8.0
10 b	N0572766	7	166	64	92	6	14, 7	7.6	9.6
11 b	N0572766 (D3)	8	182	64	105	5	12,355	6.3	8.0
12 b	N0576767	6	165	74	94	5	28, 5	8.1	9.0
13	N0578768	12	136	52	17	11	7, 33	10.3	15.0
Mean		10 sites	151	45	47	7	−3, 22	5.7	9.0

The site numbers correspond to the labels in Fig. 1. N = the number of directions determined from each site or the number of site-mean directions used to calculate the overall mean. $\bar{D}$ = the mean declination; $\bar{I}$ = the mean inclination; k = the estimate of Fisher's [27] precision parameter. α_{95} = the apical half angle of 95% confidence; lat. = the latitude of the associated pole; long. = the longitude of the associated pole; dp and dm are the semi-axes of the oval of 95% about the pole [28].

[a] These site mean directions were calculated excluding the results from samples from the steep limbs of D4 folds.

[b] The overall mean direction was calculated excluding the results from sites 10, 11 and 12, which are affected by D4 movement.

zones in Glen Esk, i.e. a dip of 20° to the south-east.

Fig. 4 shows that scatter is significantly reduced when structural correction for D4 movement is applied demonstrating that the characteristic magnetization predates D4 folds in this part of the Barrovian zones. The structural corrections have a degree of uncertainty because the precise attitude of the fabric of the Glen Esk sites at the time of magnetization is not known. The directions of characteristic magnetization of samples and sites known to be affected by D4 deformation are not used in the calculation of the site-mean directions or the overall mean direction listed in Table 1.

5. A test for an anisotropy effect

There remains the problem of the effect of a strong metamorphic fabric on the directions of magnetization. Stacey and Bannerjee [19] dismiss all strongly foliated rocks as unlikely to record the palaeomagnetic field because of the probable deflection of the remanence from the direction of the inducing field. Various suggested corrections for deflection of remanence by a rock fabric incorporate low field anisotropy of susceptibility measurements [20].

This problem may be addressed using the data from the D3 structures. It can be determined if there is a statistical tendency for the characteristic magnetization to be deflected into the planes of regional foliation deformed by D3 folds, before the magnetization of these rocks. The closely grouped directions of magnetization from the D3 fold in site 11 are plotted in Fig. 3 with projections of the planes of regional foliation corresponding to each fold limb. The fold is plunging slightly to the northeast. Three samples were collected from the upper limb and three samples from the lower limb. The distribution of direction of characteristic magnetization is independent of position on the fold and of the angle of direction of magnetization with respect to the regional fabric. Unless the mean direction coincides with principal axes of susceptibility on both limbs of the fold, or the orientation of the ellipsoids is such that they produce the same deflection in the same direction, the haematite schists at site 11 are not affected by detectable anisotropy of remanence. A similar exercise applied to site 8 in Glen Lethnot leads to the same conclusion.

6. Mean palaeomagnetic direction and pole position

The site-mean directions, overall mean direction and associated pole positions and statistics are given in Table 1. The mean direction from sites 1 and 3 were calculated excluding the results from the steep limb of the D4 folds. The results from sites 10, 11 and 12 were not used to calculate the overall mean direction because of the probable involvement of these sites in D4 deformation and the uncertainty of applying a structural correction at these sites.

The palaeomagnetic pole calculated from the overall mean direction is plotted in Fig. 5 with Palaeozoic poles from the Grampian Highlands which include poles from the Aberdeenshire Gabbros [21], the Newer Granites [22,23] and late Silurian/early Devonian volcanics [24,25]. The Barrovian pole falls on the Aberdeenshire polar wander track between poles derived from sites with estimated K-Ar biotite chrontour ages of 468 m.y. and 448 m.y. The average biotite K-Ar age for the Barrovian zones in the vicinity of the



Fig. 5. The pole from the Barrovian zones (*BZ*) is plotted with other poles from the Grampian Highlands. Poles labelled with numbers are from [21]; the numbers being K-Ar chrontour ages in m.y. *C* is the Comrie pole [22] *A* and *GF* are the Arrochar and Garabal Hill Glen Fyne poles [23]. *LP* is the Lorne Plateau volcanic pole [24] and *GC* is the Glen Coe volcanic pole [25].

sampling sites is 455 m.y. and the average Rb-Sr biotite age is 450 m.y. [4]. The Barrovian pole seems to be consistent with the Aberdeenshire polar wander track in that the pole falls in a place consistent with a K-Ar biotite age of 455 m.y. However, the blocking temperatures of magnetization in the Barrovian haematite schists which have 455 m.y. cooling age is 600° to < 695°C, while the blocking temperature of typical Aberdeenshire Gabbro with this same cooling age is 450–570°C.

7. Discussion and conclusions

The haematite schists of the lowest kyanite and staurolite Barrovian zones in the Grampian Highlands of Angus, Scotland, have a characteristic magnetization with blocking temperatures greater than 600°C and less than 695°C, and coercivities greater than 200 mT. There is no detectable effect due to anisotropy of remanence. This magnetization is not folded by D3 folds but is folded by D4 folds. The associated pole position falls well away from the Newer Granite and Devonian volcanic poles and lies on the older part of the Aberdeenshire polar wander track. Specifically, it falls between poles derived from estimated biotite K-Ar chrontour ages of 468 m.y. and 448 m.y. This is apparently consistent with the average K-Ar biotite age of 455 m.y. for the relevant part of the Barrovian zones.

The very high blocking temperatures observed in the haematite schists in fact suggest that the magnetization is of chemical rather than pure thermal origin. The peak temperatures expected in the lowest kyanite and staurolite zones are 580–600°C. If the magnetization was a pure TRM the maximum blocking temperature would only be a few tens of degrees Celsius higher than this either because of geologically slow cooling [15] or long periods of static heating [26]. The magnetization is most likely to be chemical in origin acquired during or shortly after the peak of metamorphism after D3 deformation.

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